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Machine Learning for Wireless Communication Channel Modeling: An Overview

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Abstract

Channel modeling is fundamental to design wireless communication systems. A common practice is to conduct tremendous amount of channel measurement data and then to derive appropriate channel models using statistical methods. For highly mobile communications, channel estimation on top of the channel modeling enables high bandwidth physical layer transmission in state-of-the-art mobile communications. For the coming 5G and diverse Internet of Things, many challenging application scenarios emerge and more efficient methodology for channel modeling and channel estimation is very much needed. In the mean time, machine learning has been successfully demonstrated efficient handling big data. In this paper, applying machine learning to assist channel modeling and channel estimation has been introduced with evidence of literature survey.

Keywords Machine learning · Channel modeling · 5G · MmWave · Mobile communications · Indoor/outdoor communication systems · Regression · Deep neural network

1 Introduction

After a century's efforts on radio communications, state-of-the-art mobile communications, from cellular communications to vehicle-to-vehicle (V2V) communication, good channel modeling and fast channel estimation are essential to high-bandwidth physical transmission technology. The state of art of channel modeling, which will be reviewed in Sect. 2, is defined as a fundamental part of the physical layer in communication systems. The purpose of channel modeling is to extract and develop channel state information (CSI) such as fading, delay, Doppler and azimuth etc, from channel impulse response (CIR). In the classical channel modeling, the general expectation maximization (EM) algorithm is used to estimate the CSI of each multipath components (MPCs) by applying space-alternating generalized expectation-maximization (SAGE) algorithms which is an extension of

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the EM. With the usages of increasing number of users, applications and MIMO, SAGE is becoming more complex and other sophisticated and computationally techniques are required [1].

In other words, channel modeling uses mathematical parameters to obtain the effect of the channel medium. These effects cause the transmitted signal to be either destructive or constructive during the propagation. Where the main focus of this paper is how machine learning will be used in channel modeling. Channel modeling can be facilitated by scholastic or deterministic methods. The stochastic modeling method is based on the statistical distributions of the channel parameters and can be either narrowband or wideband. While deterministic such as ray tracing is base on Maxwell equation. In the narrowband, the impulse response within a delay bin can result in flat or frequency non-selective fading. Whereas in the wideband, fading focus on each ray of channel response individually and can be a dispersion of signals, or frequency selective fading. Where machine learning techniques can be applied to both narrow and wide band. The general wireless channel model is shown in equation below and other MmWave channel can be found in [2]. Equation (1) shows finite-difference time-domain (FDTD). The time-variant CIR is expressed for frequency selective channel as:

$$y(t) = u(t) * h(\tau, t) = \int_{-\infty}^{\infty} h(\tau, t) u(t - \tau) d\tau \quad (1)$$

where $y(t)$ is the received signal, $u(t)$ is the transmitted signal, $*$ is the convolution sign and $h(\tau, t)$ is the delay spread function with respect to delay and time.

Then by applying machine learning algorithms to model the wireless channel by using such a Bayesian techniques to estimate the channel state information such as Doppler spread f_d and delay τ from a dataset $x_1, x_2, \dots, x_n$.

$$L(n/x_1, x_2, \dots, x_n) = \prod_{i=1}^n f(x_i/n) \quad (2)$$

$$[\hat{f}_d, \hat{\tau}] = \arg \min_{\sigma, \alpha} -L(f_d, \tau) \quad (3)$$

$$\arg \min_j error = \sum_{i=1}^n [y_i - \hat{y}_i] \quad (4)$$

Channel estimation is a mathematical predicting of the natural propagation of the signal that helps the receiver to approximate the affected signal, which plays an important role for highly mobile or dynamic channels. Fifth generation (5G) with massive MIMO, higher data rate, quick handover, and channel modeling become more and more complex than the traditional stochastic or deterministic methods. In this sense, researchers and academia looking forward to more effective methods that has less complexity and more accuracy. Emerging machine learning technology that will be reviewed in Sect. 3 supplies a new direction to process big measurement data and traffic data. The 3rd Generation Partnership Project (3GPP) standards group currently working to involve machine learning for the 5G era. The network data analytics function (NWDAF) is a new part of the 3GPP standardization where its main concern is the data and the format of the analytics information that it uses [3]. As Fig. 1, machine learning applied to wireless communication channel modeling

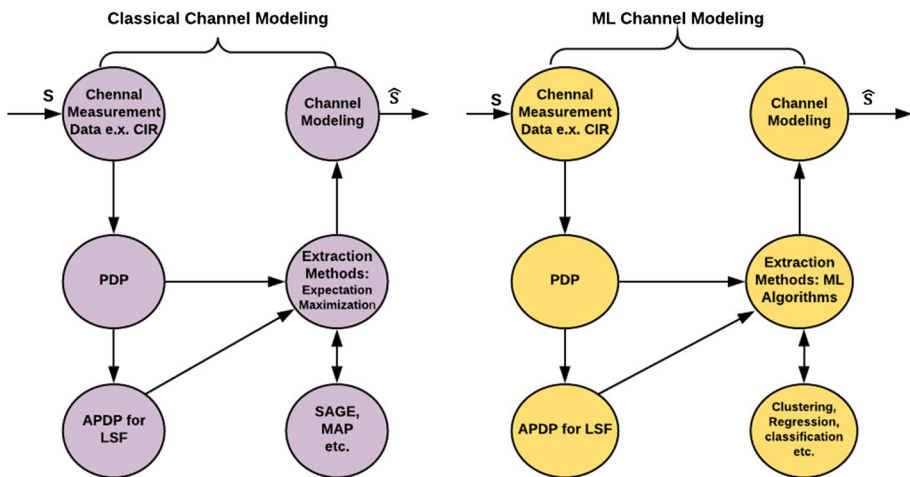


Fig. 1 A general visualization explaining the difference between traditional and new wireless channel modeling

is expected to improve the derivations of channel models and contribute to more effective channel estimation. Consequently, machine learning for wireless channel modeling is not only best fitting a model with appropriate parameters but also keeping the model updated to obtain more precise operating of wireless communication systems.

Figure 1 demonstrates a general compression of the classical wireless communication channel modeling and the ML wireless channel modeling. Both of them start with have CIR that is obtained from the sounders and then power delay profile (PDP) can be determined by taking the absolute square of the CIR to determine the small scale parameters. If the target is to estimate large scale fading (LSF), average the power delay profile (APDP) [4]. Then, using classical estimation method such as SAGE to extract the channel parameters. On the other hand, involving and applying machine learning to extract the CSI and then applying estimation techniques using method such as regression, classification and clustering etc. to estimate the channel parameters that will lead to a better performance and lower complexity. Channel estimation is a mathematical predicting of the natural propagation of the signal that helps the receiver to approximate the affected signals. [5] suggests using deep learning to identify and classify the modulation nodes, improving the interference alignment and locate the optimum routing path.

In this overview manuscript, channel modeling is needed to be understood well to be able to combine the machine learning algorithms in which is in Sect. 2. Then, Sect. 3, shows ML usages in channel modeling including estimate the wireless channel by machine learning algorithms. ML applications, challenges and open research topics in channel modeling explore in Sect. 4, such as cellular and V2V. Lastly, a conclusion is shown in Sect. 5.

2 Channel Model

The main intention of the channel modeling in the area of cellular systems is to support basic physical layer communication. Channel modeling [6–9], intends to mathematically describes the multipath components (MPCs). Channel modeling can be implemented by

either Deterministic or Stochastic manner, to the reorientation of the propagation waves through a channel and how these waves get affected by the channel surrounding. Then, a mathematical representation describes the channel impulse response in the time domain or by channel transfer function in the frequency domain. In parallel with the measurement and theoretical research, the simulation tool is building to reduce the complexity and improving the performance.

2.1 Background

Wireless channel is the logical connection between the transmitter and receiver where the wave is propagated carrying the information. That wave reaches to the receiver after experiencing effect due to channel fading and notice. The modeling methods can be either deterministic or stochastic where all of these lead to a precise channel impulse response (CIR) with different complexity. Stochastic can be either empirically or simulated build to reconstruct CIR after predicting the received signal fading. Next, the data received has to be extracted to obtain the data and its feature from the received signal and establish a relationship between large scale or small scale parameters [10]. One method to do that using space-alternating generalized expectation-maximization (SAGE) algorithm that uses maximum a posteriori (MAP) channel estimation method [11]. Instead of applying that classic method, machine learning method such as Principal Component Analysis (PCA) will do the job with less complexity.

Channel modeling of wireless communication [10] can be represented mathematically by the time variant CIR impulse response $h(t, \tau)$ in the time domain or in the frequency domain by using time-variant channel transfer function $H(t, f)$. The channel impulse response can be obtained by sounding the channel periodic pulses with PN sequences or pulse at the transmitter while on the frequency domain the channel can be sounded with multi-tone signals such as VSG. Both time and frequency domain are describing the behaviors of the signals where Fourier transform is used to obtain the transfer response in the frequency domain. The mathematical models can include parameters such as carrier frequency, shadow, phase, Doppler, delay, distance, path loss and others where most of these parameters are varied randomly [12].

Fading is identified as the time variation of the received multipath components MPCs power that went through the medium channel effects where it can be categorized into two parts as Small-Scale Fading and Large-Scale Fading. Propagation in wireless channel models results in either large or small scale fading base on the variations of the signal. Large-scale fading can be either path loss or shadowing due to shadowing by blockers that are larger than the carrier wave. While small-scale fading is due to amplitude variations due to multipath time delay or Doppler spread.

Before applying machine learning techniques, we need to understand the fundamental of the wireless channel and how to model it. There are two types of channel modeling, which are Deterministic and Stochastic Channel Models. Other may divide the wireless communication channel models into three catalogs as geometry based deterministic models such as Ray tracing, non-geometrical stochastic models as the empirical models and geometry based stochastic models (GBSMs) which is a combination of deterministic and empirical models. Deterministic channel modeling: The main idea of this is to reproduce the communication by using Maxwell's equations. The CIR will be produced using such as ray tracing model with geographical data. The disadvantage of this approach is every single location has different surfaces which require different design especially with vehicles

that pass through a different scenario in a period of time and it demands a highly precise geographical database. In this approach, all objects such as buildings, cars, trees, and others are divided into a small slot that gathered to in the ray tracing. There are other models such as finite-difference time-domain (FDTD). The time-variant CIR is expressed for frequency selective channel as:

$$y(t) = u(t) * h(\tau, t) = \int_{-\infty}^{\infty} h(\tau, t) u(t - \tau) d\tau \quad (5)$$

where $y(t)$ is the received signal, $u(t)$ is the transmitted signal, $*$ is the convolution sign and $h(\tau, t)$ is the delay spread function with respect to delay and time.

$$h(\tau, t) = \sum_{k=1}^{N(t)} a_k(t) e^{j(2\pi f \tau_k(t) + \varphi(t))} \delta(\tau - \tau_k(t)) \quad (6)$$

Furthermore, α is the amplitude, K is the MPC at time t and τ is the delay and is φ the phase shift. While stochastic channel models approach can be divided into two classifications, geometry-based stochastic models (GSCMs) and non-geometry based stochastic models. The Non-geometry models can either a narrowband stochastic channel models or wideband stochastic channel model [10]. Where the narrowband stochastic model usually focuses on the characterization of the fading statistics as well as the Doppler spectrum. Whereas wideband stochastic model is typically focused on the received power, delay, angle of departure and arrival and Doppler shift, etc. However, the geometry base stochastic models which are similar to the non-model but with a simplified ray tracing. Most of the current MIMO channel modeling use this model. The difference between wideband and narrowband is the amount of the bandwidth however in narrowband, the delay has no much affects base on the relation $T_m \approx B^{-1}$ and $u(t - \tau_n(t)) \approx u(t)$ but with wideband all of these approximations are not valid. T_m is the multipath delay spread, B is the bandwidth, τ_n is the delay and $u(t)$ is the low pass signal. Moreover, narrowband characterization is limited to model the path loss and coherence time. While wideband used to characterize the channel impulse response to investigate the delay parameters since the narrowband measurements aren't able to resolve the individual multipath components in the time domain [7].

2.2 Large-Scale Fading

Large scale fading usually due to the object that shadow the signal or longer distance between transmitter and receiver. Path losses and shadowing can cause a large fading to the transmitted signals that degrade the signal strength. However, the channel modeling has not been well studied. Most of the published path loss models follow the general path loss model in the below equation.

$$PL(\text{in decibels}) = [\text{Free space loss at 1M}] + 10n \log(d) + X\sigma \quad (7)$$

where $X\sigma$ is the normally distributed random variable mean and σ standard deviation plus other unknown losses added to this parameter during fitting the model which in our analysis would be investigated more. Moreover, the receive signal strength varies over distance r as $\frac{1}{r^2}$ for free space or $\frac{1}{r^4}$ for ground reflection [13]. Additionally, the relationship between path loss and transmitted power is showing as:

$$PL[dB] = P_t[dBm] - P_r[dBm] + G_t[dB] + G_r[dB] \quad (8)$$

where PL is the path loss, P_t is the transmitted power, P_r refers to the received power, G_t and G_r is the transmit and receive antenna gain, respectively

2.3 Small-Scale Fading

Small-scale fading is due to the constructive and destructive of the multipath components MPCs at the receiver side which usually leads to fluctuations over small distance or small interval of the signal envelope [10]. Besides, Small-scale fading is used to determine the link level performance based on bit error rates and average fading. Usually, the amplitude distribution of the received signal is modeled as a Rayleigh distribution for the non-line of side where Ricean distribution is used for the Line of side. Where Rayleigh fading models are the most common statistical model for the random amplitude. Some classes of such fading are summarized as follows:

- Flat fading: This type of fading are recognized as the variation of the signal amplitude and it's signal bandwidth is narrow compared to channel bandwidth. Usually, flat fading occurs if the signal width pulse or symbol period is larger than the delay spread $T_s \gg \sigma_\tau$ or the signal bandwidth is less than the channel bandwidth $B_s \ll B_c$.
- Frequency selective fading: Frequency selective fading is due to the time spread of the transmitted symbols within the channel [14]. Frequency selective fading is also known as wideband channels and occurs if the signal width pulse or symbol period is less than the delay spread $T_s < \sigma_\tau$ or the signal bandwidth is larger than the channel bandwidth $B_s > B_c$. Frequency selective fading can be overcome by two methods. For example, for narrowband bandwidth transmissions if frequency selective fading occurs at the transmission frequency, then the entire signal will be lost unless two methods applied. The first one is by transmitting a wide bandwidth signal or signal spectrum as CDMA where any dips or fading only occurs in a small loss of the signal power instead of a complete loss. The second method is to separate the transmission into small bandwidth carriers or chops such as in OFDM transmission. In both cases the loss is small where this information can be recovered.
- Slow Fading: Slow fading is happening when the channel impulse response changes repeatedly within symbol duration or the channel variation is smaller than the baseband signal variation or if the Doppler spread is low. Slow fading occurs if the coherence time is large relative to the symbol period delay required by that application or standard $T_s \ll T_c$ or the signal bandwidth is larger than the channel bandwidth $B_s \gg B_c$.
- Fast Fading: Fast fading is happening when the channel impulse response changes rapidly within symbol duration or the channel variation is faster than the baseband signal variation. Fast fading occurs if the signal width pulse or symbol period is larger than the coherence time of the channel $T_s > T_c$ or the signal bandwidth is less than the channel bandwidth $B_s \ll B_c$. Moreover, fast fading occurs when the Doppler spread is high [10].

2.4 Distortions in Channels

There are two kinds of distortions, time-dispersion (inter-symbol interference) or frequency-dispersion that the wireless channel can get affected by as shown in the below subsections.

2.4.1 Time Dispersion

When a signal is propagated through the medium, the signal experiences a distortion affected by reflections, scattered and diffracted by objects where the signal arrived at different times called time dispersion. The different time between the first and last multipath components called delay spread. Time dispersion can cause inter-symbol interference ISI. This dispersion is due to the signal bandwidth larger than the coherence bandwidth while frequency selective fading is due to time spread in the time domain. Usually, the sideband signals mitigate the dispersion than the narrowband [14, 15].

RMS Delay Spread RMS delay spread [10] can be obtained by taking the square root of the second central moment of the PDP. Usually, rms delay spread values are in order of microseconds for outdoor environments and nanoseconds in the indoor radio channel. Once the RMS delay spread obtain from PDP's, the maximum RMS delay spread, the average RMS delay spread, and the standard deviation RMS delay spread costs can be achieved. While The mean excess delay is the first moment of the PDPs.

Maximum Excess Delay Maximum Excess Delay is the time delay value that counts after the MPC's energy become lower than X dB below the maximum multipath energy. It is also called excess delay spread.

2.4.2 Frequency Dispersion

Frequency dispersion is due to mobility while time selective fading is due to frequency spread in the frequency domain. This has to do with the effect of receiver mobility. If the signal changes rapidly enough in comparison with the coherence time, then there is no distortion.

Coherence Bandwidth Coherence bandwidth is a method that used to characterize the channel in the frequency domain. It is a statistical measure of the range of frequencies over which the channel can be considered flat. Coherence bandwidth also is the difference between frequencies over which two frequency components have a strong potential for amplitude correlation [10].

Delay Spread and Coherence Time Both Delay spread and Coherence bandwidth used to describe the time dispersion of the channel in a local area. They don't offer information about the time-varying nature of the channel caused by relative motion of transmitter and receiver. It is measured of spectral broadening caused by mobility, the time rate change of the mobile radio channel.

One of the ways to compute Doppler shift: fd Doppler spectrum has components in the range of $f_c - f_d$ to $f_c + f_d$. If the symbol period of the baseband signal is bigger than the coherence time, than the signal will be distorted, since channel will keep changing during the propagation of the signal.

Doppler Spread (B_D) Doppler spread is just the frequency shift of a narrowband channel due to the mobility of vehicles or the surrounding environments. Doppler spread can be derived as a function of effective mobility through different environments [16]. Where the square root of the second order central moment is used to characterize the empirical Doppler spread B_D from the frequency domain recordings of the Rx. Doppler spread is defined as the maximum Doppler shift: $f_m = v/l$. If the baseband signal bandwidth is much less than the Doppler spread then effect of Doppler spread is negligible at the receiver and this would be a slow fading channel characteristics. In practice, the Doppler-variant

spread function is used to examine the Doppler spread. Doppler spread can be obtained by the inverse Fourier transform of the spreading function $S(\nu, \tau)$ with respect to time where the spreading function is just the Fourier transform of the time-variant impulse response $h(t, \tau)$. The Doppler function is shown as:

$$B(\nu, \tau) = \int_{-\infty}^{\infty} S(\nu, \tau) e^{-j2\pi f \tau} d\tau \quad (9)$$

Moreover, it is a good indicator of the time selectivity (RMS Doppler spread) and is related to the coherence time. According to [17] Doppler spread is studied where its effect due to the higher mobility of the vehicles nodes which cause Doppler shift (DS) of the receiving node in the carrier frequency. Mitigating the Doppler effect will improve the throughput and the quality of service (QoS) as well. Also, the effect of Doppler frequency shift on the carrier frequencies is happening in different modulation with the same percentage and the SNR is degradation on the OFDM system due to the Doppler shift and other reasons such as frequency offset and phase noise.

3 ML Usage in the Wireless Communication Channel Modeling

Artificial intelligence (AI), particularly machine learning (ML), is widely studied to enable a system to learn of intelligence, predict and make an assessment instead of the needs of humans. Switching the traditional channel modeling to machine learning channel modeling still in its early stage. One of the main issues in current communication is to accurately predict the channel parameters, whereas using machine learning techniques could enhance the prediction and reduce the complexity. ML can be used to predict and estimate the wireless channel parameters and examine large and small-scale fading including parameters such as path loss, delay path loss exponent, carrier phase shifts, Doppler spread and random variable that explains the large scale fading. Zhang et al. [18] used multiple of machine learning algorithm such as KNN and Random Forest to estimate the path loss model from a dataset and showed results that prove the accuracy with small estimation errors.

Estimation these parameters is required at the receiver. In ML, there must be a different way than the previous methods such as minimum mean square error (MMSE) estimation or maximum a posterior probability (MAP). The new ML methods estimate the channel parameters using different methods base on their categories as shown in Table 1. OShea et al. [19] used deep learning for estimating the carrier frequency offset (CFO) and Ben-Hur et al. [20] showed how to predict the transmitted signals using deep learning

Table 1 An overview of machine learning algorithms in wireless channel modeling

Category	Tasks	Algorithms
Supervised learning	Regression, classification, bayesian inference	Neural networks, decision tree. Logistic regression, K -nearest, SVM, Gaussian regression
Unsupervised learning	Clustering, feature extraction, dimension reduction	K -means, spectrum clustering, PCA
Reinforcement learning	Policy learning	Q-learning

techniques such as neural network other than the classical method where existing receivers estimate the parameters and then recover the data using estimation. Whereas with deep learning, the channel state information (CSI) are estimated and recovering the transmitted signals directly. He et al. [21] used neural network methods such as learned denoising-based approximate message passing (LDAMP) network to estimate and learn CSI and solve the limited number of frequency chains in cellular or V2V systems from training data. Jing et al. [1] used model-based method using Cramer-Rao lower bound (CRLB) to predict the CSI in deeper neural network. During the data processing, the noise threshold has to be identified where any PDP close to the noise floor will be identified during the data processing. Specifying the noise threshold to extract the CIR is an essential thing. The traditional ways to overcome the noise from the CIR and signal higher than the threshold is considered acceptable and choosing the noise threshold is sensitive since removing all noise means getting rid of part of the signal and reducing the noise level leads to having some noise in the signal itself. Machine learning methods are capable of choosing the accurate threshold. OShea and Hoydis [22] presented several novel ideas of applying deep learning to the end-to-end communication systems such as autoencoder and redesigning that communications in a single process. Where the autoencoder helps decreasing the block error rate (BLER) by 1–7 times in scenarios that have Rayleigh fading. Moreover, the same author presented in [23] other channel modeling challenges that reach the complexity level such as blockage, atmospheric effects, handover, beam direction, MIMO and it is the time for machine learning to get evolved. Raj and Kalyani [24] proposed a unique way to train deep learning of channel modeling without any assumptions which improve the system performance. Authors used a back propagating stochastic approximation method to overcome the corrupted signals due to hardware, fading and improper schemes etc. The cross entropy loss function used to measure the performance of the system as shown below.

$$L(\mathbf{s}_{1:N}, \hat{\mathbf{s}}_{1:N}) = -\frac{1}{N} \sum_{n=1}^N \sum_{s_i \in s_n} s_i \log(\hat{s}_i) \quad (10)$$

where s is the input symbol and $\hat{s}$ is the estimated output.

Table 1 shows that machine learning can be divided into three main categories that can be applied to the wireless channel modeling in general. These categories are supervised, unsupervised and reinforcement learning.

3.1 Supervised Learning

For the purpose of learning and predicting, common methodologies to realize ML are training, testing, and validating wherein training part, learning from the data while the testing method, a trained model is used for predicting such as channel modeling parameters [25]. Supervised learning techniques require input, target and training data to create a model that is used for estimation. If we have a sample space consist of X_i and output label space y_i where $i \in 1, 2, \dots, N$ then by using a machine learning algorithm $\hat{A}$, which is a function that map the input values to the labels where it helps for future predicting. To measure the quality of the mapping, a loss function is used and see [26] for more detailed. Usually, the supervised learning can divide into two main subjects which are the regression model, the classification learning. The regression method is considered continuous values whereas the classification is a discrete value. Both of these two subjects are useful in estimation the channel parameters such as the path loss component n and the large-scale random variable

ψ_{σ} . Also, it can be used to estimate the throughput. The error can minimize if the data is sufficient in the supervised learning.

3.1.1 Classification

The general goal of classification is to map a categorical label to each data in the provided dataset [27]. Once the radio wave propagates through the channel between the transmitter and the receiver, that wave faces physical properties. These physical effects cause the transmitted signal to be either destructive or constructive during the propagation which leads to fading. This type of fading is classified into two part. Large-scale (LSF) and small-scale fading (SMF). Usually, in LSF, the signal loses its strength due to distance in the case of users are far away from each other or due shadowing by environments obstacles. When the signals or multipath gets destructive or constructive, small-scale fading occurs and usually, small-scale fading describes the random phase and amplitudes of the different multipath echoes where these components cause attenuation in the signal strength. Moreover, these measured data will lead us to emulate a new statistical (Empirical) channel model using ML techniques to design an optimized wireless channel modeling as in [28]. Moreover, Shiva et al. [29] used multi-task classification for estimating the AOD by using sigmoid function. for the neurons functions.

3.1.2 Deep Neural Networks (DNNs)

Deep Neural Networks (DNNs) is a mini hot topic of artificial neural networks (ANNs) [30]. DNNs is the mapping of the input to the output by a graph which is a branch of deep learning algorithms. In [31], the authors recommend to involve the convolution neural network (CNN) to the 3GPP channel model. In [32], the CNN techniques used to predict the speed and direction angle using a sequence of images as an input. By using a dataset that is produced from channel modeling simulation or measurement, the deep neural network is inspired by the human brain and can be implemented by mapping the input data to the output in term of the diagram where each vertical nodes are called layer. The first layer usually from the input data and every node is connected to every node in the following layer, the last layer is the predicted or the response value and other layers between are called hidden layers. Furthermore, the more layer added refers to the depth in the network [33].

Mao et al. [5] mentioned that deep learning is an ideal technique that is capable of to solve or reduce the complex problems. Furthermore, by applying DL, the wireless network applications can reach higher estimation precision and decrease the design complexity. Authors of [29] used the neural network to optimize the estimated parameters with two hidden layers. The parameters that were estimated by DNNs are received signal strength (RSS) and delay over angle of departure (AoD) bins. Usually, the dataset is splitted into two subsets which are training and testing the model. Training the dataset is used to train the algorithm for the DNNs model and then later the tested data use to evaluate the final model [34]. Mao et al. [5] recommends using deep learning for the incomplete or erroneous data to be reconstructed as if human brain is able to recognize letters, numbers, faces etc. that is not clear enough. This method could lead the effort to increase the accuracy of link budget due to channel fading, interference and mobility.

Figure 2 shows a feedforward DNNs that can be applied to channel modeling to estimate the channel parameters such as delay, azimuth and fading. Moreover, it can be used to estimate the link budget for instance, path loss parameters. The number of neurons in each layer can

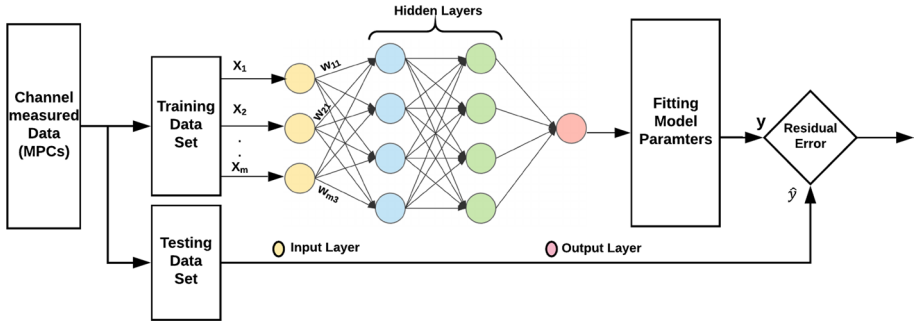


Fig. 2 Applying deep neural networks in wireless channel modeling

be shown in a vector $L = (L^{in}, L^1, \dots, L^h, \dots, L^n, L^{out})$ where L^{in}, L^h, L^{out} are the number of nodes in input, h th hidden. Let (x, y) the measurement data of the wireless channel such CIR is used in train part using back propagation algorithm or Newton algorithm [35] and the DNN algorithms to estimate the weight parameters to build the model channel. Then, the test data used to examine the model. lastly, the residual error is measured to obtain the accuracy of applying the DNN algorithm to the wireless channel modeling. Equation 10 shows the output of the DNN where w_i is the weight of each node which is needed to be estimated and n is the number of nodes. The number of output layer is usually base on the target and deciding number of hidden layers and number of neurons in each hidden layer are still an open research topic where few or more neurons leads to underfitting and overfitting. Authors of [20] used DNNs algorithm to predicts the transmitted signal and then modeling the estimated data to recover the transmitted signals. Besides, Ye et al. [36] addressed that DNNs is capable to learn the channel model characteristics of the wireless channel that get affected by fading.

$$h(t, \tau) = \sum_{i=0}^{L-1} a_i(t) \delta(\tau - \tau_i(t)) \quad (11)$$

The time-varying impulse response of the channel is shown in Eq. (10) and after applying the APDP to it using $APDP(\tau) = \frac{1}{N} \sum_{i=0}^N |h(\tau)|^2$. Then, the output of that equation are fed to the structure of the DNN. To predict the Y_i values from the training data $X_1, X_2, \dots, X_N$

$$Y_i = f\left(\sum_i w_i x_i + b\right) \quad (12)$$

where b is obtained by the training data and one way to determine the bias is using root mean squared error (RMSE) and by using iteration of the neural networks, the b value can be reduced [25].

The raw data of the received and transmitted signal are collected as a training data and the model is trained and then the loss L is obtained.

$$\arg \min_j L = \frac{1}{N} \sum (\hat{Y}(k) - Y(k))^2 \quad (13)$$

To minimize the error between the estimated and the received transmitted signal, sum of square residual error of sample i is used.

$$\arg \min_j e_i = \sum_{i=1}^n [y_i - \hat{y}_i] \quad (14)$$

The dataset of the wireless channel is supposed to divide into two fragments as in Fig. 2. The fragments are training, and testing which help to produce a perfect model. Other research use training, validation and testing [37]. The selected model will be trained and validated on the dataset then will be tested using the unseen data as showing in Fig. 2. Usually, the train part takes 60% of the data set and 25% for the validation and 15% for tasting the model [38, 39]. The training process can be repeated several time and then taking the average as [1] repeated the training process ten times.

Ye et al. [36] purposed a method to train the receiver by computing the loss function and using the DNN. The input of the DNN is the received signal y and the received signal pilot y_p (for the time-variant) that leads estimate the channel without estimating the channel by itself. The transmitter and channel can be trained and estimated as shown in Hao et al. work. Hackeling [40] purposed DNNs to improve based automatic modulation classification (AMC) technique. Authors had revealed that using DNNs is powerful for channel modeling and different than the traditional channel modeling. Huang et al. [41] used DNN to estimate the channel of MIMO systems using the direction of arrival (DOA). Authors trained the DNN using data of different channel scenarios that confirm the better performance than the conventional methods.

3.1.3 Support Vector Machine (SVM)

Support vector machine (SVM) that is derived from learning theory is a method to analyze data for classification analysis or regressions. SVM was invented by Ben-Hur et al. [20], to identifying dataset using separating hyperplane. Linear regression is suitable for a complex problem and it was extended to SVM. Linear regression is the fundamental of the SVM. Zhao et al. [42] showed how SVM method can help predicting path loss which made it easier and more reliable than the classical prediction. Zhao et al. used a method to estimate path loss by training PL data and then built a model that is capable of predicting the unmeasured data. Moreover, SVM can be used to classify by finding the hyperplane that differentiates the LoS and NLoS.

3.1.4 Regression Algorithms

Regression is one of the main methods that is used in machine learning where the regression models are a learning mechanism based on a dataset from prior measurements or simulations. After finishing the learning processes, the models can be used for predicting other parameters. Authors of [28] applied three regression models support vector, linear and DNNs regressions then, made a compression to control the high-speed channel modeling errors. Where all of these techniques help reducing the wireless channel modeling complexity and need to be investigated more. Shiva et al. [29] used the regression algorithms to estimate the RSS over delay bins. In Sect. 4.4.1, regression will be applied to path loss.

3.2 Unsupervised Learning

Usually, it is hard to have a large amount of labeled data which cause people to switch to use the unlabeled data which is known as the unsupervised learning. Using machine learning techniques such as unsupervised learning to reduce the complexity, improve the measuring performance and distributing the wireless cellular tower. Unsupervised learning is useful in the case when the data isn't efficient where the hidden variables can be obtained using methods such as Bayesian learning. Clustering is one of the main methods of unsupervised learning wherein channel modeling is used in gathering multipath components (MPCs) with similar parameters behavior which can improve the process of the time-variant channel model precision. These MPCs has similar parameters such as angle of arrival (EOA), angle of departure (EOD), delay (τ), azimuth angle of arrival (AOA), azimuth angle of departure (AOD), and power level will be in the same cluster [43]. As illustrating in Fig. 3, MPCs performing in the cluster phenomenon where part A shows MP power delay profiles (PDPs) that have similarities in one cluster and part B was adopted from [44] that shows least-square regression of PDPs within clusters. Figure 3 is also explaining the clustering method in the wireless channel modeling where each color represents a single clustering showing either the LoS and NLoS or multipath components.

3.2.1 Gaussian Mixture Model (GMM)

The Gaussian mixture model (GMM) is a method of clustering techniques where it's used to implement channel multipath clustering. GMM gives the option to mix the relation of data to clusters. By comparing k-means which is a clustering algorithm to GMM in data distributed, in K-means there is no mix where a single input only belongs to one cluster (Hard clustering) whereas in the GMM an input can belong to different cluster at the same time with a different degree of similarity (Soft clustering). GMM is more flexible than k-means in terms of reducing the complexity. Using such as simple principal component analysis (PCA) is an example of the big data algorithms to model a channel modeling where the PCA uses an orthogonal transformation to a correlated dataset to uncorrelated data set [45]. Moreover, showed the MPCs assumed to consist of several Gaussian distributions in the GMM technique that can be applied to channel multipath clustering and the likelihood of this distribution is

$$L(X; \Theta) = \sum_{i=1}^N \log \sum_{k=1}^K (x_i | \mu_k, \Sigma_k) \pi_k, \quad (15)$$

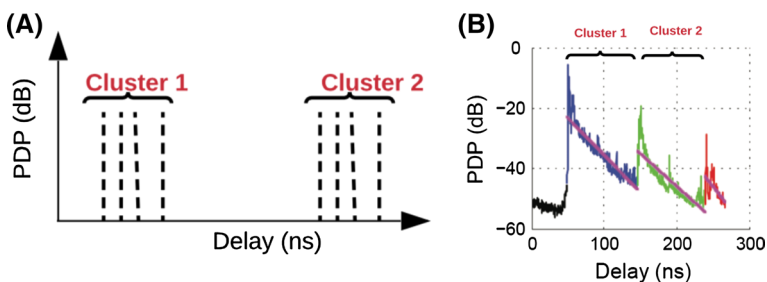


Fig. 3 Applying clustering techniques to the channel modeling **a** MPCs delay spread. **b** Clustering in delay simulation [44]

The theta parameter Θ is the assembly of parameters such as μ_k , Σ_k and π_k . Where π_k is the prior probability of a class. x_i is the channel multipath, μ_k is the mean and Σ_k is the covariance matrix and π_k . Then, parameters of the GMM methods are estimated to solve the likelihood function by the expectation maximization (EM) algorithm.

$$\hat{\Theta}_{ML} = \underset{\Theta}{\operatorname{argmax}} L(X; \Theta), \quad (16)$$

3.2.2 K-means Algorithm

Mohri et al. [26] used the K-means algorithm to identify the clusters and then this algorithm iteratively groups cluster the MPC to minimize the Euclidean distance between the data until it becomes a cluster and then keep iteration to reach the convergence. This algorithm has been used in [35] to cluster the multipath. During the channel modeling, if there are L number of cluster and each cluster consists of $X_1, X_2, \dots, X_n$, of multipath. By using the centroids $C_1, C_2, \dots, C_L$, which is the cluster center in random locations in space and a point in high dimensional vector space. Then, by using the measured data set and for each multipath X_i find the nearest centroids C_i and assign each multipath X_i to every cluster j. Then by using the below equation, Euclidian distance D_E can be obtained to cluster the transmitted signals base on their behavior.

$$\underset{j}{\operatorname{argmin}} D_E(X_i, C_j) \quad (17)$$

where the $D_E = \sqrt{\sum_{i=1}^k (x_i - y_i)^2}$ for continuous features and for the categorical variables, Hamming distance is useful as $D_H = \sum_{i=1}^k |x_i - y_i|$.

The second step of the algorithm is recomputing each cluster centroid position $j = 1, 2, \dots, L$ and that can be done by taking all data inside that cluster and averaging them to obtain the new centroid j. Then, these steps kept iterations to readjust the centroid location until it converged.

As showing in [46], after collecting the measurement data and estimation the channel parameters, several studies have noticed these parameters such as the delay (τ), azimuth angle of arrival (AOA), azimuth angle of departure (AOD), elevation angle of arrival (EOA), elevation angle of departure (EOD) usually have the same cluster. Ko et al. [47] used a clustering algorithm to sort the MPCs with similar behavior such as delay and angular characteristics. Additional, authors of [48] used K-power means clustering algorithm to automatically clusterize the MPCs based on multipath component distance (MCD). MCD is used to enhance the clustering performance and [49] use other distance measures and compare them. Li et al. [45] is also proposed kmeans-based clustering algorithm to classify the delay of MPCs and keep iteration the MPCs until the distance of the slimier data is minimized. The number of an optimum cluster can be calculated using Kim-Park method [50] or other use numbers of a cluster as a random variable as in COST 259 models [51]. The total MCD between the i th and the j th multipath is shown as

$$MCD_{ij} = \sqrt{(|MCD_{Rx,ij}|)^2 + (|MCD_{Tx,ij}|)^2 + (|MCD_{\tau,ij}|)^2} \quad (18)$$

$$MCD_{\tau,ij} = \frac{|\tau_i - \tau_j|}{\Delta\tau_{max}} \cdot \frac{\tau_{std}}{\Delta\tau_{max}} \quad (19)$$

K-power means (KPM) is a clustering algorithm that is used to assign a clustered index to each MPCs which is beneficial to minimize the MPCs base on their power.

$$\arg \min_j KPM = \sum_{l=1}^L P_l * MCD(x_l, C_j) \quad (20)$$

L is the number of MPCs, P_l is power of each MPC as $l = 1, \dots, L$ and parameter vector x_l represent delay and other angular parameters. Then, iteration is used until it converged to its own clustering.

Ma et al. [52] proposed a channel modeling method to exploits the feature and structure obtained from the channel impulse response using an unsupervised method called principal component analysis (PCA). The CIR data obtained from measurement that was conducted in teaching building of BUPT, China. Zheng et al. [53] and Huang et al. [54] are one of the researchers that proved the ML techniques can improve the wireless channel modeling. Authors proposed a tracking algorithm based on maximization estimation and *K*-power means algorithm for the time-variant wireless channel and showed a good performance by clustering and estimating channel parameters using nonlinear least square regression method. Li et al. [55] purposed estimate the MPCs by space-alternating generalized expectation maximization algorithm and used clustering methods such as MCD tracking algorithm and the *K*-power means algorithm.

Figure 4 illustrates the processes of machine learning techniques usages in wireless channel modeling. The block diagram explains how MPCs are estimated and then clustering using cluster algorithm such as *K*-power means. Data later fed to the clustering partitions to be identified the observations into k clusters then the output shows the channel parameters. cluster validation is also recommended during the clustering processes by using methods such as Calinski-Harabasz index and the Davies-Bouldin criterion to generate significant results [56].

In, telecommunication propagation the MPCs are cluster into group base on LoS and NLoS and the strength of the received signal energy. classical clustering such as *k*-means and spectrum clustering for the above causes. Ye et al. [33], compared two unsupervised methods, *K*-means, and GMM to model using clustering techniques in the area of wireless channel multipath. Jiang et al. [1] applied Cramer-Rao lower bound (CRLB) to switch the CSI inference problem to extract the channel modeling parameters.

3.3 Reinforcement Learning

Another branch of Machine Learning, known as reinforcement learning (RL), is to learn from experience when the prior information is unknown, or to interact with dynamic environments. RL is used to learn a system to solve complex problems such as Markov Decision Process (MDP) which usually called model-free based method or model-free methods

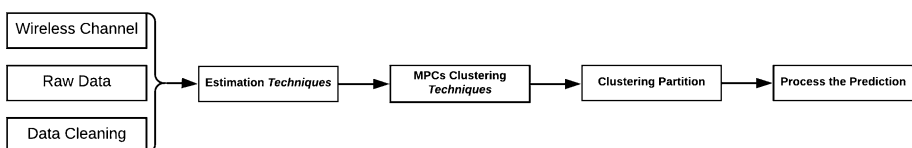


Fig. 4 Flow chart of channel cluster modeling processes

such as Q-learning that does not learn a system [57]. RL starts with learning, updating and then pursuing the action to maximize a reward. Markov Decision Process (MDP) is the foundation of the RL where the goal of an MDP is to find an optimal policy to maximize the decision [58]. Markov Decision Processes are widely used as a methodology for modeling sequential decision-making in environments with probabilistic dynamics or dynamics of uncertainty.

In the channel modeling of the wireless sensor networks (WSNs) consist of resource-limited devices such as power transmitting. Static decision commands may lead to inefficient energy usage. For example, a node sending data at fixed transmit power without considering the channel conditions will drain its energy faster. Thus, by using MDPs will help to optimize the problem. The MDP model allows a stable design of dissimilar objectives, such as minimizing energy consumption and maximizing sensing coverage. Ultra-reliable millimeter-wave communication with higher frequencies the more loss occurs due to blockage and weakness of the propagates signals. Bennis et al. [59] proposed a risk-sensitive reinforcement learning to optimize the beamwidth and the transmit power of cells. The optimal policy can be obtained after having the optimal value function to select the action that maximizes the value in the next state.

Reinforcement learning is all about trying to understand the optimal way of making decisions/actions so that we maximize reward R such as Q-learning. RL can be applied to issues that have been a while that the wireless system dealing with such as reinforce the traffic flow and handover where an action the machine has to do. Applying RL in channel modeling will lead to better performance such in power control and handover [60].

$$\begin{aligned} \pi^*(x_k) = & \operatorname{argmax}_{a_k} \{E[r_{k+1} | x_k, a_k] \\ & + \beta \sum_{x_{k+1}} p(x_{k+1} | x_k, a_k) V^*(x_{k+1})\} \end{aligned} \quad (21)$$

A useful algorithm that helps to find the optimal value function is called the value iteration algorithm.

$$\max_{x \in S} |V^{l+1}(x) - V^l(x)| < \delta \quad (22)$$

where l is an iteration counter. The above equation can be used to compute optimal policy using value iteration which is the most effective and widely used technique to resolve infinite time horizon discounted MDP. This method has many advantages such as quick convergences especially if the state space is too large and easy to implement. One of the Algorithms that represent RL is using Iteration which usually repeat the optimization until the optimal state value function $V(x)$ converges. The converges occurs once the maximum value difference between two iterations is less than a specific threshold. For the policy iteration: we keep repeating improving until it converges.

Machine learning techniques have been a major method for the problems that deal with routing such as in adaptive routing, shortest path, and multicasting routing. One of the ML techniques to solve these issues is Q-learning is. Q-learning is one of the reinforcements learning that is used to control the routing follow [61]. Usually, Q-learning is a model-free method which doesn't learn a system and can handle the routing by using stochastic transitions and rewards. Q-learning calculates the policy that maximizes the expected value to maximize the overall reward. Q-function is used to solve the MDP problem in cases when the information of the environment isn't needed. Ye et al. [62] mentioned that Q function estimates the expected reward when taking an action in a state such as pursuing a handover and power control problems. RL can be valuable in blockage and selecting beam issues

where there will be a reward if the beam of the signal is selected successful and if there is a blockage, a negative reward is incurred and reelecting the link base on the training mode is required. To the best of our knowledge, there is a rear works that have been done on applying reinforcement learning on wireless communication channel modeling [58]. Deep reinforcement learning (DRL) as mentioned in [5] can be used for cognitive radio network (CRN) control to sense the free channel.

3.4 Massive Wireless Channel Modeling Data Analytics

The channel modeling datasets involve massive amount of data, and make the data analysis to be time-consuming. Effective machine learning and feature extraction can alleviate the thrust from the size of data sets, and the complexity with comparable functionality can be substantially reduced. Authors of [63] explore complexity reduction techniques to improve the signal efficiency. The first technique is using distributed optimization algorithms for instance, alternating direction method of multipliers (ADMM) and primal/dual decomposition. Both of these methods are used to decouple the statistical learning issues into small sub-issues. The second technique is used for incomplete data and to perform that, online or active learning is needed.

Other types of classification reduction are dimension reduction which is used to reduce a high dimensional space without losing any useful information. Methods such that are liner projection and nonlinear projections methods. An example of the linear projection methods is principal component analysis (PCA) and for the nonlinear methods are iso-metric mapping (ISOMAP) and locally linear embedding. Zhang [64] went over an inter-disciplinary study of using massive channel modeling data and used cluster-nuclei based channel model and matching the propagation signals to machine learning algorithms using measurement data.

Data Mining is used to learn rules and relationships that help to build a model to predict automatically from the updated data. Data mining can be obtained from too many resources where this overview paper scope is wireless channel modeling which becomes a Data Mining. Data is useful for the scientists and during this decade, wireless communication devices have had grown by a factor of one thousand [15, 65]. Since the measurement of channel modeling will produce big data with a large of Gigabits especially with V2V and V2I. MIMO is one of the major things that switch the channel modeling to be a data mining. Wang et al. [66] recommended to use the beneficial of data mine to model wireless channel modeling.

Usually, the parameters of a channel model are generated by measurements or simulations of the propagated signal through a channel where that signal gets disturbed by fading, blocking and distance which leads to MPCs. The highest MPCs is the strongest link and from there, channel parameters can be obtained to create a dataset. During the previous wireless generation where SISO with lower bandwidth, the amount of data is not sufficient enough to be used for accurate prediction. Whereas with the usage of MIMO and higher bandwidth, data can be used to estimate instead of taking so much measurement and leads to support the accuracy of wireless communication modeling. Thus, channel modeling can be considered as a sort of data mining.

The data is needed for establishing a channel model where the features and structures are obtained for the empirical data. Generally, wireless communication including cellular, V2V and others can be a data mining since it's geographically distributed, producing data continuously with updated versions of the same parameters [67]. Data mining

methods used in Machine learning as other algorithms to build models that predict our needs. Authors of [52] believe that the combination of data mining and machine learning methods and huge dataset, channel modeling parameters can be extracted and controlled. Hackeling [40] have confirmed that deep model reduces the channel modeling complexity by allowing DNN algorithms to learn complex functions. Moreover, some techniques can be applied to the data analytics unit such as confirmatory data analysis (CDA) which uses statistical techniques to find out whether hypotheses of the data set are true or false. Also, exploratory data analysis (EDA) technique can be applied to the data analytics to obtain the relationships and patterns in the given data [68].

Model Validation Data used to build the proposed models have to be validated using different methods to evaluate the goodness of fit (GoF) such as R-Square and the root mean squared error (RMSE). R-Square is used to measure the success of the fitted model where it ranges from -infinity to 1, where value one considers the optimum regression model to the collected data. RMSE (root of MSE) checks the accuracy of the wireless channel propagation predication where it measures the differences between the predicted and observed model where zero value indicates the fit is optimum [10].

We may use the statistical parameters P-value, F-value or alpha level to achieve the significant of the predicted or the used model. Furthermore, these parameters can be used to validate the significance and check the accuracy of the proposed model. As far of our current knowledge, there has not been a clear vision of how to use the channel data mining to improve the channel modeling agreeing with [44].

4 Applications of Machine Learning in Channel Modeling

The propagation channel of communication systems fundamentals is focusing on channel modeling particularly for a new technology era such as machine learning. Channel modeling is so wide either in cellular or Vehicle to Vehicle (V2V) channel model systems. The irregularity and complexity of the wireless channel lead to solid ways to achieve accurate models where more trails are always required to accomplish the precise results mainly with new technologies. Where the primary focus of this chapter is to go over Machine learning in different channel modeling applications.

Machine learning is needed to examine the channel modeling and focusing in regarding large scale and small-scale fading where fading refers to the time changes of the MPC's power. The large-scale fading (LSF) explains the main characteristics of the channel such as path loss, shadow, angular spread, and delay spread. Moreover, LSF cases will be examined the relationship between the path loss and the separated distance between the Tx and Rx in different environments such as suburban. Where the small-scale fading case will go over mainly the relation between the Doppler spread and time coherence. These studies are needed due to the lack of real measurements which will allow developing a reliable V2V communication. Generally, large-scale parameters are chosen base on the mobility, shadowing, delay spread and angular spread. while the small-scale parameters are measured based on power, Doppler spread, an angle of arrival and departure of the MPCs.

4.1 MmWave Channel Model in High speed Applications

Investigations on the vehicle to vehicle (V2V) or high-speed train channel modeling, would allow other applications to interchange data to make the communication securer by using robustness detection scheme to avoid jamming attacks [69]. Getting machine learning involved in a measurement from simulation or campaign will provide us with unbiased results. Machine learning algorithms investigate the feature of wireless channels deeply in MmWave. Moreover, these measured data will lead us to build a new statistical (Empirical) channel model to design an optimized a communication with high-speed applications. Machine learning is known to improve the performance and reduce the complexity. Lu [28] applied ML techniques for high-speed channel modelings such as support vector and deep neural network (DNN) regressions. By using a dataset from prior measurements and designs, ML methods would be applied to reduce the errors.

During the past decade, vehicle-to-vehicle communication has gained academical and industrial intentions where the vision is to have communicated vehicles for safety and other purposes. Today, there are a couple of challenges to create wireless communication such as V2V and that is due to latency which will slow the interchange of data from sensor or camera between connected vehicles and complexity. Base on the literature, the only way to overcome these issues is using Machine learning especially with future technologies that supports high data rate [70]. The usages of higher frequency will assist V2V to interchange information to create what is known as a connected vehicle due to the amount of bandwidth and high data rate. Another issue with V2V is both Tx and Rx are at the same height which leads to high scattering [71]. By using machine learning algorithms, Efficiency and lower complexity can be achieved than the current V2V communication systems.

During the V2V communication, the Tx, Rx and the environment moving which lead high Doppler shifts. The channel modeling in V2V is a time and frequency variant. There is a couple of V2V channel modeling such as Two Ring Model [72, 73] and Regular-Shaped GBsMs (RS) [74] where both of them are 2D GBsMs. A third V2V channel model was proposed by [75] is a 3D two-cylinder based and later was extended by the same author into a non-stationary 3D model [76]. Too many V2V channel models in the literature but since it is a moving environment, Time-Delay Line (TDL) is a non-stationary correlated scattering and modeled based on real measurement in different places [77].

3GPP [78] has release outdoor-to-indoor car penetration loss and that can be modeled as

$$PL = PL_b + N(\mu, \sigma_p^2) \quad (23)$$

From the above equation, PL_b is the outdoor PL and that model can be applied for frequencies from 0.6 to 60 GHz.

Penetration Loss Modeling was proposed by the 3GPP, which consists of the following parameters which can be possibly used for the future V2I.

$$PL = PL_b + PL_{tw} + PL_{in} + N(0, \sigma) \quad (24)$$

where PL_b is path loss from the urban macro or urban micro models and PL_{tw} is the loss due to building penetration. PL_{in} is losses inside building and σ is the standard deviation. Furthermore, these parameters can be modified base on the location and versions update of the model [78].

Since we have the three main challenges to implement ML in V2X communication systems, i.e., the lack of ML vehicular channel models, the ML algorithms, and usage of deterministic or stochastic such as the empirical models. The current issue with ML V2V is the lack of accurate ML channel model. Couple characteristics in the channel modeling

need to be investigated with ML algorithms such as radio signal strength (RSS), delay spread, Doppler spread, phase shift and bit error rate. These characteristics usually used to improve the performance of the V2V communications and ML will reduce the complexity. V2V and V2I is a non-stationary channel where the medium channel changes during short time. That change is due to the mobility of one or all transceiver or the environment by itself such as other moving vehicles cause time variant. These actions also lead to delay spread and that is noticeable due to very high delay resolution [4].

By applying machine learning to the vehicle to vehicle (V2V) or vehicle to infrastructure (V2I) to the wireless propagation channels to measure the parameters such as large scale and small scale fading such as path loss, shadow, delay parameter. Moreover, how to predict these parameters form a row data. One of the ways to predict these parameters is using Bayesian models such as the hidden Markov models that can use the current state data to predict the other parameters. Other methods may be used for estimating the channel model such as neural networks based on dataset.

4.2 V2V/V2I Channel Modeling

This section describes some machine learning techniques toward the challenging V2V/V2I channel modeling. Channel modeling of the V2V/V2I is different than the cellular system due to multiple reasons such as the mobility of Tx, Rx, surrounding environments and being low comparing to the cellular systems. Also, vehicle obstruction, traffic congestion, and tunnel lead to shadowing and extra delay spread that is not an issue in the cellular systems. Moreover, the amount of the Doppler shifts is significantly higher due to mobility which causes the channel to be non-stationary. Base on the previous reasons channel modeling is different and be either Deterministic and Stochastic Channel Models [10]. Deterministic channel models as a 3D GBSM such as two-ring model whereas the stochastic is done using the empirical models such as the TDL model. Other V2V channel models have been proposed for the stationary and non-stationary channel for different scenarios can be found in [79, 80]. Here, a CIR of a wideband directional model for time-variant V2V is shown below [81]. The channel impulse of the the 3D GBSM models and other channel models can be found in a recent publish book [6].

In V2V, All machine learning categories can be applied similar to the cellular systems and an of using the reinforcement learning in autonomous decision maker—the agent—has to determine actions to execute in order to achieve an objective such as selecting best beam. One of the reasons for using the dimensional reduction is estimation would be easier especially with mobility and in V2V. V2V/V2I in the 5G will be a data mining and get the privilege of that using machine learning techniques will improve the autonomous. ML also can predict the vehicular direction or vehicular route by using networking with data afforded including things like roadmap or driver history which helps prediction the traffic follow. Its useful to solve the near-far problem by making a decision of the amount of energy that supposes to be transmitted.

Applying machine learning to the vehicle to vehicle (V2V) or vehicle to infrastructure (V2I) is still an open subject and how to alternate from the traditional ways of measuring the channel empirical model or the ray-tracing methods.

4.3 MmWave Channel Propagation Model in Cellular Systems Environments

Similar channel model to the previous sections, channel modeling of cellular systems can be achieved using Deterministic and Stochastic Channel Models. The geometrical stochastic models such as the empirical models can be used to identify the large-small scale model. In this section, path loss models and how to apply machine learning techniques will be exploited. Moreover, there will be a couple of recent large-scale model to obtain the path models that consider the shadowing effects to be one of the parameters. PL describes how the mean received power varies with the distance between the transmitter and receiver where it's due to attenuation due to shadow fading between (SF) Tx and Rx. PL is a measure of the channel range and quality in the link budget. By matching the channel modeling formulas to the statistical analyses such as the path loss.

MmWave emerges as a new technology opportunity for 5G new radio. Most of empirical path loss models derive from the data collection and then compared with theoretical models such as free space and two-ray model.

$$PL(d) = PL(d_0) - 10n * \log_{10}\left(\frac{d}{d_0}\right) + X_{\sigma}(d); d \geq d_0 \quad (25)$$

where $PL(d_0)$ is the path loss at reference distance d_0 , n is the PL exponent which is an environment's dependent that is determined by field measurements, and X_{σ} is the large-scale fading about the mean power which follows normal distribution with zero mean and variance σ^2 as X_{σ} follows $N(0, \sigma^2)$. $PL(d_0) + [10n\log]_{10}(d/d_0)$ can be used as the mean of the path loss $\mu(d)$ and can be estimated and the large-scale fading X_{σ} can be also estimated using empirical measurements or regular estimation using a distribution. To obtain the large-scale fading, the effect of the small-scale fading is supposed to be removed from the data set which usually done by averaging the set over time sample or other methods.

ML still relies on relatively old-fashioned techniques such as logistic regression, SVM, decision trees, K-NN, naive Bayes, Bayesian modeling, ensembles, random forests, signal processing, filtering, graph theory, gaming theory, and many others to extract the channel modeling information. Involving a machine learning techniques to estimate the path loss parameters still rely on traditional techniques such as regression and Bayesian theory. The PL candidates are the alpha-beta-gamma (ABG) model, the close-in (CI) model and Close-in with frequency dependent exponent [82].

4.3.1 Alpha-Beta-Gamma (ABG) Model

ABG model is the current 3GPP 3D model and values may change based on the base station location. Academic and industrial are trying to come up with a single path loss model for each scenario.

$$PL^{ABG}(f, d)[dB] = 10\alpha\log_{10} + \beta + 10\gamma\log_{10}\left(\frac{f}{1GHz}\right) + X_{\sigma}^{ABG} \quad (26)$$

ABG model is used to measure how PL get increased with distance and β is the optimized offset in dB. Also, γ is the PL variation over frequency in GHz and X_{σ}^{ABG} is the fading (SF)

in dB. Since there are three parameters, 3GPP mentioned that ABG PL model always has a lower shadow fading standard deviation than other PL models [83].

4.3.2 Close-in (CI) Model

This model is usually for LOS and NLOS for all UMi, UMa, and InH as well by using close-in reference distance base on Friis' law. The general version of the close-in model is [84]:

$$PL^{CI}(f, d)[dB] = FSPL(f, 1, m) + 10n \log_{10} \left(\frac{d}{1m} \right) + X_{\sigma}^{CI}, \quad (27)$$

The above equation is a path loss model where the parameter FSPL is just the free space model in dB, is the path loss exponent (PLE) to show PL varies with multipath propagation distance and d_o is the reference distance which set to 1 m since there is no shadowing in the first meter and simplifies the equation as well. The FSPL can be obtained using $FSPL(f, 1, m)[dB] = 20 \log_{10} \left(\frac{4\pi * f * 10^9}{c} \right)$ Note: FSPL at 1 m and c is the speed of light. Close-in path loss model is more robust and provides better performance than the ABG path loss model [85].

4.3.3 Close-in with Frequency Dependent Exponent

This model was proposed in the 3GPP meetings where it depends on how high is the frequency. Close-in with frequency dependent is an extension of the CI model with frequency dependent [86]. The general model of the CIF model shown as [87].

$$PL^{CIF}(f, d) = FSPL(f, 1, m) + 10n \left(1 + b \left(\frac{f - f_o}{f_o} \right) \right) \log_{10} \left(\frac{d}{1m} \right) + X_{\sigma}^{CIF} \quad (28)$$

Note: b is a parameter that captures the slop and it would be positive if the both PL and f increase. f_o is the reference frequency.

Sun et al. [88] compared the above path loss model for both CmWave and MmWave in different scenarios such as UMa and UMi. All of them showed a good prediction with large data where the CI model was the most suitable for outdoor cases due to the close-in free space reference. Whereas, CIF has a better performance for the indoor environments due to its small standard deviation values.

4.4 ML Channel Estimation

Due to the increasing complexity due to a large number of operations and large measurement, researcher tend to perform machine learning to enhances the channel modeling. There are multiple of machine learning estimator methods that used to predict the channel parameters such as maximum likelihood estimation, Minimax Estimation, and regression that use least Square Estimator, the minimum mean square estimator [28]. These methods will obtain the channel state information (CSI) from the detected signals. Authors of [29] used regression to estimate the the received signal strength over AoD bins. Riihijarvi and Mahonen [25] have applied several of machine learning methods to estimate the wireless network using data

measurements. The authors use Gaussian regression and random forest to predict where reducing the operational cost and enhancing the performance have been confirmed in this paper. Wang et al. [89] used machine learning algorithms such as regression to predict the receive signal strength using beam training. This approach could raise the awareness of autonomous vehicles accidents using their locations using machine learning and ray tracing. Burghal et al. [90] used linear and logistic regression to solve the switching band and band assignment problem then compared the results with different structures of neural networks. DNN can be used also to estimate the path loss as shown in [91].

ML can estimate the large-scale fading using supervised learning such as Regression model. By using data to predict and infer the main parameters of the PL such as n and the large-scale fading σ . σ is from the error term in the general path loss equation and since the error is assumed to be i.i.d. where its mean is zero and variance is σ .

4.4.1 PL Estimation Using Regression

Regression is a machine learning under the supervise category [28]. Regression use LSE is used to minimize the square of the error between the observed responses in the dataset and to predict the most accurate model [18, 25]. LSE can be divided into two classes: linear (ordinary) least squares and nonlinear least squares where both of them don't require any prior knowledge.

Path loss (PL) considered the main goal for the link budget and by using a likelihood expression for a normal distribution to estimate the PL and the large-scale parameters. Where the data set is assumed to be Gaussian distributed due to a large amount of data that converge to normality and since large-scale fading is already assumed to be Gaussian. If we assume having N sample data point where $(X_1, Y - 1), (X_2, Y_2), \dots, (X_n, Y_n)$. By using the general form of multiple Regression to estimate the PL variables is:

$$y = f(x) + \epsilon \quad (29)$$

where y is the dependent variable (PL) and sometimes it called the output response, x is the independent variable and the ϵ is the theoretical error. However, since we have more than one sample the function of x would shown as:

$$f(x) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p \quad (30)$$

The estimated of the response variable y and minimum goal of erroneous follow:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \dots + \hat{\beta}_p X_p \quad (31)$$

$$\arg \min_j \sum_{i=1}^n e_i^2 \quad (32)$$

Then, estimating the mean and variance can be shown below.

$$\hat{\alpha} = (X^T X)^{-1} X^T y \quad (33)$$

Using the above estimate, we can infer the variance σ^2 ;

$$\hat{\sigma}^2 = \frac{1}{L-1} (y - X\hat{\alpha})^T (y - X\hat{\alpha}) \quad (34)$$

Machine learning estimation would be involved to predict the channel modeling parameters that would help to reduce the signal error $e(n)$. Also, Bayesian or ANNs methods can be

used to predict the output parameters of the channel. Moreover, the minimum mean square estimator is a channel estimator which predict the modeling parameters that maximize the likelihood.

4.5 Challenges and Open Issues in Channel Modeling by Machine Learning

Applying machine learning to enhance or to enable channel modeling is still in the infant stage. We summarize a list, surely not an exhaustive list, in the following:

- Channel Modeling under overfitting and underfitting with massive data that leads to estimation erroneous [48, 92].
- Analysis massive data of channel modeling with advanced machine learning algorithms [21].
- Applying machine learning techniques to MmWave channel challenges such as Blockages [15].
- Interference such as co-channel interference (CCL) and inter-symbol interference (ISI) [14, 15, 33].
- Securing the data during channel estimation in V2V [69].
- Power control and handover where decision supposed to be built based on previous states or results and making a decision using reinforcement in the current stage such to execute a handover [93].
- Accurate channel models with dynamic behavior with mobility [94].
- Switching between dual bands system (MmWave $\iff$ Microwave) [95].
- New standardization suppose to include machine learning and other data formats.

Solving these issues require more investigation and powers form researchers in academia and industrial. Resolving these issue is essential for future smart cities and artificial intelligence life.

5 Conclusion

This paper presents an overview of theoretical framework toward machine learning problems for future wireless communication channel modeling. Even with the high demand to emerge machine learning to the field of modeling, there are still some challenges and block thoughts that need to be solved. This article presented the 5G in channel modeling domain and summarized the current progress of applying ML to channel modeling. The main difference between the current LTE and 5G is the usage of data and how to involve machine learning to it. This paper went over some useful methods in machine learning in the field of wireless communication such as regression analysis which were used for data fitting to predict the channel modeling parameters. Moreover, how classification identifies the input data to be mapped to the output which can be used to predict the traffic flow. Reinforcement learning is all about trying to understand the optimal way of making decisions/actions so that we maximize reward R such as Q-learning. Applying RL in channel modeling will lead to better performance such in autonomous vehicles.

Furthermore, other channel modeling scenarios that reach the complexity level such as fading, atmospheric effects, handover, beam direction, MIMO, suggest the time for machine learning to get evolved. Machine learning techniques show the viability proceeding with

estimation the channel parameters and extracting the channel information as showing in estimating the path loss parameters using regression and clustering techniques.

Additionally, this paper presented several open research topics that required more efforts. So far, there are so many investments and time from academia and industry have been invested in machine learning of the wireless channel modeling and we still have a long way to overcome the challenges and achieve the precise modeling. Involve machine learning to the wireless channel modeling is still on the preliminary phase and such that work will lead and guide scholars to achieve that goal. The scientific explanation of using machine learning in wireless communication channel modeling has been shown with evidences and reasons through this journey, while mobility introduces further technological opportunities. As a conclusion, applying effective predicative ML methods can reduce the complexity and increase the accuracy than the regular channel modeling. Involving ML techniques to the wireless channel modeling becomes more efficient as the number of measurements is reduced as a good generalization abilities.

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